# Assumptions & Decisions

Project rules: assumptions, decisions, taxonomy, formats. The rules register of the Ocean Opportunity Radar, rendered from PROJECT_RULES.md at f091af8. Times are UTC.

The project's one register of rules. Follow it in every session. Sections:

- **A, assumptions:** standing rules applied the same way to every record.
- **D, decisions:** dated decisions, each with the time (UTC), the rule, the reason and the records affected.
- **T, taxonomy:** signal types, sources, confidence levels, verdicts, group tags.
- **F, formats:** dates, IDs, countries, amounts.
- **Q, open questions.**
- **K, known inconsistencies.**

This file folds in ASSUMPTIONS.md (bf4547c, merged to main as e0568b2) and keeps its numbering. A1 to A27 and D1 to D20 are the same rules under the same ids. New items extend the numbering and nothing is renumbered.

How the register is used:

- An open case is asked in a batch (`QUESTIONS_B<n>.md`). Live pages keep the current rule until the case is answered (A31).
- An answer becomes the next D number.
- Every change-log entry cites an A or D id (A32).
- `python3 -m radar.consistency` checks the stores against this file before delivery.
- `python3 -m radar.rules_pdf` renders this file as the "Assumptions & Decisions" chapter, `docs/method/assumptions_and_decisions.pdf`.

This file is re-issued after every question batch.

Sources and dates: the thesis files, the code, EVAL.md, README.md, MODULES.md, METHOD.md, ARCHITECTURE.md, tasks/cloud/*.md and `git log`. Times are UTC commit times. Counts are the ones the repo states; where none is written, the entry says so. `thesis.yaml` is `config/thesis.yaml` (blue economy, Finland); `baltic` is `config/thesis_baltic_blue.yaml`; `health` is `config/thesis_health_turku.yaml`. A rule with no thesis named is the same in all three.

## A. Project assumptions

**A1. One source of record for a company's facts.** For a Finnish company the PRH/YTJ register sets legal name, founding date, industry code and description, website, status and town. The declared list, Hilma and CORDIS may insert a company first, but `prh` re-reads every company that has a business ID and replaces those fields with the register's non-null values; a NULL never erases. Business Finland, tax registers, OpenAlex, the web archive and the ecosystem dataset only match companies that exist (the dataset inserts an unmatched company tagged `ecosystem`, outside the feed). The AI stitcher writes no company field, only checked signals (A17). The repo holds no ranking among Business Finland, Hilma, CORDIS and the declared lists; they do not compete, because only the register overwrites. Town comes only from the register address (visiting address, else postal; abroad stays NULL). For Norway the register supplies founded, code and status and the declared name stays. **Example:** the declared list names Arctic Fry Oy; the register then gives founded 2020-10-23 and code 03220 (EVAL.md). **Where:** `radar/emit.py upsert_company`; `radar/adapters/prh.py _refresh`; `radar/build.py` PHASE; `radar/town.py`; `radar/nordic.py import_cohort`.

**A2. What a "startup" is (wide intake).** A company enters intake when its PRH company form is 16 (limited company) or 14 (cooperative), never a sole trader or partnership; its industry code starts with 03, 26, 27, 28, 29, 30, 582, 62, 63, 71 or 72, or is a code under `industry_codes`; and it has `min_signals: 1` startup signal of type first_sales, first_hire, eu_project, eu_grant, research_output, bf_funding, startup_notice, equity_event or public_funding, never the register entry. It enters on that signal's date. Intake says nothing about sector fit: the row reads "Thesis fit: no public evidence yet" until evidence appears. `health` uses prefixes 21, 26, 32, 62, 63, 72 and has neither 582 nor 03. **Example:** Sireness Oy (72100) entered on 2025-09-29, rank 20 of 103 new 2025 companies in family 72 (EVAL.md). **Where:** `intake:` in each thesis file; `radar/intake.py`; `/method#intake`.

**A3. Listed companies are left out.** A legal name with Oyj or Abp among its last two words is left out of the feed and the intake tier; the footer and `/method#left-out` print how many. **Example:** "Solar Foods Oyj" is left out, "Oyjala Boats Oy" is not (`tests/test_round3.py`). **Where:** `radar/store.py` `LISTED_FORM_WORDS`, `is_listed_company`; `radar/intake.py`.

**A4. Human-health practice is not health technology (`health` only).** Companies whose industry code starts with 86 (clinics, physiotherapy, medical specialists) are left out of the feed and the intake tier; 86900 is off its code list. The blue files exclude no code prefix. Now 30 companies on `health`: 1 main, 29 weak (README.md). **Example:** a clinic under code 86xxx never appears; code 32500 still does. **Where:** `feed_exclude_code_prefixes` in `health`; `store.excluded_code`; `/method#left-out`.

**A5. Strong versus weak feed row.** A row is strong (main list) only when the stored record shows at least one reason: the industry code is on the thesis list, a thesis word is in the registered line of business, a thesis word is in the source's own text (Hilma notice title, CORDIS title and objective, BlueInvest profile text; our own `detail` annotations are never read), or a thesis word in the legal name is corroborated by a record from a different source family (A6). Every other row is weak and stays in the closed Weak list. A name word alone never admits a company. On 17 Oct: 122 companies, 30 main, 92 weak (README.md). **Example:** code 30110 (ship building) alone is strong; the word "marin" in Sub Marine Store Oy backed only by a PRH tax-register record is weak. **Where:** `store.explain`, `REASON_IDS`; `/method#precision`.

**A6. Corroboration comes from a different source family.** `prh`, `prh_notices`, `prh_intake`, `prh_recent` and `tax_registers` are one family, the PRH registry, so a PRH record never backs another PRH record. A CORDIS, Hilma or Business Finland record does. **Example:** a name word plus a PRH notice is weak; the same word plus a CORDIS project is strong. **Where:** `store.SOURCE_FAMILIES`, `CORROBORATION_RULE`; `/method#corroboration`.

**A7. Consultancy and research codes are never enough alone.** A code in a `code_alone_weak` family is a hint only: the row stays weak unless the line of business, the source text or a corroborated name word also backs it. Families: 711, 712, 721 in `thesis.yaml` and `baltic`; 702, 721 in `health`. A dot is ignored (NACE 71.12 is 711). Ship building, water and aquaculture codes are unchanged. **Example:** a company registered only under 71121 reads "Weak: industry code 711xx alone; a consultancy code needs a sector word". **Where:** `code_alone_weak` in each thesis file; `store.code_alone_family`, `explain`.

**A8. Word matching.** A thesis word matches a whole word, or the start of a longer word when the thesis word has 5 or more characters; text is case-folded; `keyword_stop_words` blocks known false friends even as prefixes. **Example:** `aalto` would prefix-match a surname, so `aaltonen` and `aaltola` are on the stop list. **Where:** `radar/fit.py` `MATCH_RULE`, `MIN_PREFIX`; `name_keywords`, `keyword_stop_words`; `/method`.

**A9. Confidence levels.** High: the source names the company or its business ID directly and states the date. Medium: matched by exact legal name or website domain, date stated by the source. Low: matched by name keyword or industry code only; shown, never counted in rankings. Register discovery never exceeds medium for a name word and is low for a code alone; the stitcher is high only when the business ID or legal name and a day or month are on the page. Declared research counts in the backtest only at medium or high. **Example:** a PRH refresh of a known company is high; a new registration found by the code 39000 is low. **Where:** `radar/emit.py` docstring; `store.CONFIDENCE_RULES`; `radar/adapters/prh.py`; `/method` Confidence rule.

**A10. Readiness levels come only from the sponsors' frameworks, and unknown is never Low.** Technology = Aalto TRL 1-9, business = KTH BRL 1-9, investment = Investment Readiness 1-9 read from the ten Investment Potential dimensions, impact = the Ocean Impact Navigator category. Each level is a printed evidence-class rule over public records; no record in a class means no level, shown as "no public evidence". A BlueInvest profile's stated TRL wins over the rule TRL. We built no questionnaire scores. **Example:** Arctic Fry Oy: business readiness 2 of 9 from the register entry; technology and investment "no public evidence". **Where:** `radar/frameworks.py` `BRL_MAP`, `TRL_MAP`, `IRL_MAP`, `page_levels`; `config/frameworks.yaml`; `/method#frameworks`.

**A11. Levels become Low / Medium / High through fixed bands.** TRL, BRL and IRL: 1-3 Low, 4-6 Medium, 7-9 High; no level is unknown. Impact: a Navigator category found by rule is Medium; the dataset's own impact score is declared research and never sets a band. Thesis fit: industry code or own text High; line of business or corroborated name word Medium; legal-name word only Low; no reason unknown. The memo reads the same levels question 02 prints. **Example:** a company with a BlueInvest-stated TRL 4 is Medium technology. **Where:** `memo.bands` in each thesis file; `radar/memo.py`.

**A12. The decision table runs in passes; the first match wins.** Pass 1: the team's 10 rows, an exact match on impact, business, investment and thesis fit. Pass 2: the team's rules a to g. Pass 3 (A13). Otherwise WATCHLIST. A cell that needs a band never matches "unknown", so a missing level cannot produce REJECT / LOW FIT; that needs evidence of low fit and low impact. Pass 3 exists only in `thesis.yaml`; `baltic` and `health` stop at pass 2, and `health` sets impact to `any`. **Example:** Arctic Fry Oy: impact Medium, business Low, investment unknown, fit Medium gives WATCHLIST. **Where:** `memo.decisions` in each thesis file; `radar/memo.py` `decide`; `/method#memo-rules`.

**A13. The fallback applies only when the fit is more than an industry code.** When fit is High and impact is unknown, impact is read as Medium, implied by the sector, and rules c1 (business Low or Medium, TRL 1-3: GRANT SUPPORT), c2 (business Low or Medium: ACCELERATOR), c3 (business and investment High: DUE DILIGENCE) apply. The gate `requires_any` needs one of: a declared cohort or portfolio member, a fit reason from the source's own text, the registered line of business, or a corroborated name word. A code-only company stays WATCHLIST ("fit rests on the industry code alone"). **Example:** Viva Jets Oy reads ACCELERATOR, "because: fallback c2, impact implied by sector". **Where:** `memo.decisions.fallback` in `thesis.yaml`; `memo.fallback`, `SUPPORT_KEYS`; `/method#memo-fallback`.

**A14. Declared cohort and portfolio members get at least Medium thesis fit.** They were selected into the programme. `declared_member_band: Medium` is in all three files; question 01 reads "declared cohort company", never "no public evidence yet". **Example:** Arctic Fry Oy has no code, line of business or source text that shows fit, and still reads Medium. **Where:** `memo.bands.fit.declared_member_band`; `memo._fit`.

**A15. Countries per thesis.** `thesis.yaml`: FI, depth "registers+contracts+grants". `baltic`: FI, SE, EE, LV, LT, DK, NO, DE, PL; national depth only for FI, EE (register+grants) and NO (register); DK looks up by name or CVR number only; SE is a connector slot; LV, LT, DE, PL are "EU sources only". `health`: FI. EU-wide adapters (TED, CORDIS, EU grants, BlueInvest) run for every listed country; TED skips a country whose depth lists "contracts", so no award is counted twice, and adds only micro and small winners as new companies. **Example:** TuniSea (NO) is dated by the Norwegian register lookup; a Latvian company is reached only by EU sources. **Where:** `countries`, `country_depth`, `ted_cpv`; `radar/thesis.py`; `/method#countries`.

**A16. Companies only: person-shaped legal names are hidden.** A name of 2 or 3 capitalised words followed only by company suffixes, with at least one of Holding, Invest, Capital, Consulting, Tmi, Ky, is treated as a private person's. The feed hides it unless the company has a non-register signal (the footer counts them); the API listings, `/data`, the wide intake and the TED adapter leave it out. Words in `non_person_words` or the thesis keywords make it a business. No private individual is named anywhere; the ecosystem dataset's people fields are not imported. **Example:** "Given Family Holding Oy" is hidden; "Blue Bay Oy" is not. **Where:** `radar/fit.py` `looks_like_person`; `non_person_words`; `radar/routes/api.py`; `radar/adapters/prh_intake.py`.

**A17. AI text is only a code-checked summary of linked sources.** The stitcher's LLM proposes; code keeps an event only if the fetched page names the company and prints the date, then re-fetches and confirms the quote, the name near it and the event words. Its events never move a company between backtest groups (A22). The AI brief (`radar/briefs.py`) is written in batch, and code drops any sentence whose URL, date, number or capitalised name is not in the company's export, or that uses praise a cited title does not hold; a brief with fewer than 2 slots left is dropped. The memo and every level use no LLM. **Example:** batch of 17 Oct: 168 events passed check 1, 128 passed check 2, 29 emitted (EVAL.md). **Where:** METHOD.md sections 2 to 4; `radar/adapters/stitcher.py`; `radar/briefs.py` `verify_sentence`.

**A18. Date precision is labelled.** An event date is the day it happened, never the fetch date. A year-only source (Business Finland) is stored as 1 January with a `signal_meta` row `year` and printed "2026 (year only)"; a month-only date is stored as the last day of the month and labelled. A year-only record is never a feed entry signal and counts in the backtest as 1 January. **Example:** Sub Marine Store Oy's Business Finland record reads "2026 (year only)". **Where:** `store.is_year_only`, `_date_fields`; `radar/schema/` `signal_meta`; `/method` Sources.

**A19. Amounts are EUR from the source, with no conversion.** `amount_eur` is filled only when the source states an amount in euros. TED is kept only when its currency field is EUR and the notice has one winner; CORDIS stores the EC contribution; Business Finland stores the export's total in euros. The repo has no exchange-rate code. **Example:** a TED notice priced in SEK stores no amount. **Where:** `radar/schema/001_core.sql` `amount_eur`; `radar/adapters/ted.py`, `cordis.py`, `businessfinland.py`.

**A20. Refresh is narrow, guarded and switched off by default.** The `/data` Refresh button re-reads only the `refresh_adapters` (`prh_notices`, `prh_recent`; `baltic` defines none). It works on a copy, swaps it in with `os.replace`

only after `PRAGMA quick_check` is ok, runs one at a time (lock stale after 1,800 s), waits a 600 s cooldown, and answers 403 unless the environment sets `REFRESH_ENABLED=1`. **Example:** a failed step deletes the copy and the live store is untouched. **Where:** `refresh_adapters` in the thesis files; `radar/refresh.py`; MODULES.md.

**A21. The feed shows a company on its first counted signal in the window.** Window 90 days; one row per company; counted types are the dated public records a scout could have seen (22 types in `thesis.yaml`). Never counted: board_change (routine), capital_loss, sales_ended, employer_ended (negative), website_first_seen (a site existing is not traction). `baltic` counts only 7 types. **Example:** a first_hire in the employer register counts; a board change does not. **Where:** `geography.feed_window_days`, `feed_signal_types`; `store._feed`.

**A22. The backtest counts a signal only if the feed rule would have shown the company.** Reference date 2026-01-01 (the programme start month, declared by the team, not an entry date); a register entry counts only if the company was at most 5 years old (`stage_max_company_age_years`). Four groups that always add up to 19: found earlier by live public records, found earlier only by declared research, found later, no public signal. Declared research and stitcher events are counted apart and never mixed into the live count. Result on 17 Oct: 0, 12, 6, 1. **Example:** Arctic Fry Oy's 2020 first sales is on its page but is not counted (code 03220 is not on the list). **Where:** `radar/earliness.py` `rule_lines`; `/backtest`; EVAL.md.

**A23. Partner organisations are never shown as discoveries.** Any legal name starting A'Pelago, APelago, Redstone or Turku Science Park, and every name on `investors.csv` and `programmes_grants.csv` (71 names on 17 Oct), is left out of the feed, the radar and the intake. **Example:** Redstone-named entities never appear as startups. **Where:** `store.PARTNER_PREFIXES`, `is_partner_org`; `/method#partners`.

**A24. Declared research is labelled and never sets a level.** Events from the ecosystem dataset carry "declared research (date)" and the dataset's own confidence; low-confidence rows are leads. Its free-text summaries and people fields are not imported. It adds evidence beside a readiness level and never changes one. **Example:** the 12 cohort companies found "only by declared research" are a separate backtest group. **Where:** `radar/adapters/ecosystem.py`; `/method#ecosystem`; `frameworks.page_levels`.

**A25. Empty is empty.** A missing value stays NULL and is drawn as a hatched chip ("not in public records", "no public evidence"); no level, date or number appears without a record and a public source link behind it. **Example:** Taikoa Oy has no counted signal and its backtest lane stays blank. **Where:** CLAUDE.md; `radar/emit.py`; ARCHITECTURE.md "Empty is empty".

**A26. Public contracts: a young company's first win, or any win for a declared company.** A Hilma award signals a discovery company only if its CPV division is in `hilma_cpv_divisions` or its title holds a `hilma_keywords` word, and the company was at most 5 years old at the award; a cohort or portfolio company counts on any win. Rows whose notice has no verified public page are skipped. **Example:** a company registered 3 years before its first in-category award counts; one registered 8 years before does not. **Where:** `radar/adapters/hilma.py`; `hilma_*` keys; `geography.stage_max_company_age_years`.

**A27. Fetch etiquette and honesty lines.** Network adapters send 1 request per 2 seconds (OpenAlex 1 per second), cache raw responses under `data/cache/<name>/` and use the project User-Agent. Nothing is scraped behind a login; LinkedIn, Dealroom, Crunchbase, PitchBook and Tracxn are shown only as "connector slot, not connected" and the stitcher refuses them. **Example:** Sweden's Bolagsverket has no free API and is a connector slot. **Where:** CLAUDE.md code contract; `radar/adapters/_fetch.py`; `/method` Connector slots.

**A28. Field candidates are kept, not chosen.** Every value a source states for a company's name, founded date, industry code and description, town, website, register status or country is kept in `field_candidates` with its source, link, date read and basis, beside the value live pages show (A1). Two candidates agree under one rule per field: names by the folded name key (diacritics, case, legal-form words and punctuation removed); founded dates at the coarser of the two precisions; industry codes by digits only (NACE 10.890 is 10890); websites by host without `www.`; towns with diacritics and a bracketed note folded away; descriptions only within one language; status words mapped to active, ended, bankrupt or unknown. A company field with two or more groups of agreeing values is a conflict and goes to the next question batch. **Example:** the ecosystem dataset's founding year 2018 agrees with PRH's 2018-10-11 for Awake.AI Oy; "Piikkio (Kaarina)" and "Piikkiö" name one post office. **Where:** `radar/candidates.py` `norm`, `same`, `clusters`; `radar/schema/060_field_candidates.sql`; `python3 -m radar.candidates build` (on a store copy).

**A29. Country codes are ISO 3166-1 alpha-2.** Every stored country is a two-letter upper-case ISO 3166-1 alpha-2 code. A country name in a list is mapped through one table, and a name missing from it gives no code, never a guess. An empty country in the ecosystem dataset is read as Finland, as the dataset's own scope is Finland. **Example:** BlueInvest's "Finland" is stored as FI; "United States" as US. **Where:** `radar/formats.py` `ISO_COUNTRIES`, `country_ok`; `radar/adapters/ecosystem.py` `COUNTRIES`.

**A30. A company key is its Business ID when it has one.** A Finnish company's key is its Business ID, written NNNNNNN-C with a valid check digit. A company without one gets `<country>-<slug>` from the declared list, `name:<slug>` from a source that names it without an ID, or, when found in the Norwegian or Danish register's live window, the register's prefix and its organisation number (`radar/nordic.py` `import_rows`). A key never changes once written, so a later Business ID for a `name:` company is a duplicate to merge (QUESTIONS_B1.md, B1-12), not a rename. **Example:** Arctic Fry Oy is `3166759-6`; TuniSea is `no-tunisea`; Hailia Nordic Oy is `name:hailia-nordic`. **Where:** `radar/formats.py` `business_id_ok`; `radar/adapters/declared.py` `slug`; `radar/adapters/ecosystem.py`; `radar/nordic.py`.

**A31. An open question never changes a live value.** When no assumption decides a case (sources disagree with no rule covering it, a value contradicts an assumption, an outlier, dates out of order, a company that is both startup and investor, a possible duplicate, a link that redirects to another company, an unexpected calculation, a new rule needed), it is asked in a numbered batch (`QUESTIONS_B<n>.md`: issue ID, affected records, problem, options A/B/C with sources, recommendation and why, other records affected). The competing values are marked `PENDING_DECISION` with the issue id in `field_candidates` only; live pages keep A1. An answer becomes the next D number, applied to every matching record including those already processed, with the count changed. One issue type is never decided two ways. **Example:** B1-06 asks whether Redkik Oy is the Finnish subsidiary or the US group; until then the page shows FI and Joensuu. **Where:** `radar/candidates.py` `OPEN_ISSUES`, `mark`; `radar/stop_and_ask.py`; `QUESTIONS_B1.md`.

**A32. Every rule-driven change is logged with its rule.** A tool that changes records under a rule appends an entry to `provenance` citing the A or D id, with the records affected; a dated decision is recorded with its commit and the counts the repo states. An entry citing an id that is not in this file fails the consistency check. **Example:** each `radar.candidates build` run logs an A1 entry with its row and conflict counts. **Where:** `radar/provenance.py`; `radar/schema/061_provenance.sql`.

**A33. Nothing is delivered until the consistency check passes.** Before a ZIP, a deploy or a store swap, `python3 -m radar.consistency` runs read-only over each store and all eight checks print PASS: no PENDING_DECISION on an applied value; every signal's confidence is what its adapter's rule gives; every signal type and source has a label; every date is ISO 8601 with a precision; every Business ID has a valid check digit; every country code is ISO 3166-1 alpha-2; every amount is EUR; every provenance entry cites an id in this file. A FAIL is fixed or asked, never waived. **Example:** the blue and health snapshot copies pass 8 of 8 (17 Oct). **Where:** `radar/consistency.py`; `tests/test_consistency.py`.

# D. Decisions

**D1. 2026-10-16 20:23Z (eb1c749, implemented 98c4e1e 21:16Z). Levels use the sponsors' frameworks.**

**Rule:** TRL 1-9 (Aalto), BRL 1-9 (KTH, not 1-5), the Navigator category, and ten investment dimensions; afaa4fb (23:25Z) declined the team's own 0-20 rubrics (TASKS.md amendments; the first is headed 20:50Z, git time is used here).

**Why:** the official written brief names them.

**Records affected:** every company's readiness lines; no count stated. `readiness.business.scale` in the thesis files still says "1-5 commercial readiness"; the page prints BRL 1-9.

**D2. 2026-10-16 21:34Z (11970db). Feed precision.**

**Rule:** own-text reasons read only the source's stored text, keywords match whole words or 5+ character prefixes with a stop list, name-only matches are weak, person-shaped holding names are hidden.

**Why:** VERIFY_night2 D-N1 (f92d137): CORDIS rows entered the main list because our own annotation was read as project text, 11 rows.

**Records affected:** blue main 39 to 31, weak 55 to 62 (1 hidden); health main 37 to 34, weak 23 to 26 (commit message). API listings hid person-shaped names from 22:58Z (baad087).

**D3. 2026-10-16 22:16Z (64c39bb). The backtest uses the feed's own rule.**

**Rule:** A (A22).

**Why:** judge review item 1: "5 of 19" were counted by register-entry date alone and their codes (46900, 70200, 47279, 62100, 72100) are not on the list (`tasks/cloud/backtest_fix.md`).

**Records affected:** snapshot earlier 5, later 1, none 13 became live 0, declared 0, later 1, none 18; with local tax-register and ecosystem runs 1, 12, 5, 1 (0ef14d2).

**D4. 2026-10-16 22:46Z (0ef14d2). Counted and refused signal types.**

**Rule:** counted startup_notice, equity_event, research_output, website_milestone, eu_grant (plus first_sales, first_hire); refused board_change, capital_loss, sales_ended, employer_ended, website_first_seen, each with a written reason on `/backtest`.

**Why:** a counted type must be a dated record a scout could have seen (EARLINESS_AUDIT.md section 2).

**Records affected:** no group moved; the cohort's only new-type rows were 4 startup_notice and 15 board_change on companies the rule does not admit.

**D5. 2026-10-16 22:58Z (baad087). 72190 removed from the thesis codes; 72100 not added.**

**Rule:** `industry_codes` in `thesis.yaml` and `baltic` carry neither.

**Why:** PRH returned 0 companies under 72190 and 30 under 72100 for registrations 2026-07-18 to 2026-10-16 (an upper bound), and the audit's likely explanation is that PRH re-registered main business lines on 2026-01-01 under a new code set (EARLINESS_AUDIT.md sections 4, 5; the audit proposed keeping 72190, baad087 dropped it). Adding 72100 was proposed in 0ef14d2 and "NOT applied": it would move Sireness Oy to found earlier (live 0/0/1/18 to 1/0/1/17) at up to about 30 low-confidence feed rows per 90 days. The repo records no separate judge ruling; no commit adds 72100.

**Records affected:** Sireness Oy (3568392-1); it is reached instead by wide intake (A2). `baltic` `nace_codes` (Norway, Denmark registers only) keep NACE 72.19 and 72.10.

**D6. 2026-10-16 22:58Z and 23:14Z (baad087, 5a114ac). TOL 2025 marine codes.**

**Rule:** 03211 and 03111 replaced by 03210 (marine aquaculture) and 03110 (marine fishing) in the Finnish lists; the Baltic file also keeps 03211 and 03111 as "Estonian EMTAK code".

**Why:** the commit names 03210 and 03110 the TOL 2025 codes, and the audit ties the change to PRH's 2026-01-01 re-coding; the Estonian register has its own codes (its basic-data file carries none today, `estonia_register.py`).

**Records affected:** 12 earliness and backtest tests that used 03211 (fixtures moved to 03210); no company count stated.

**D7. 2026-10-16 23:15Z (f78f668). Wide intake.**

**Rule:** A2, shown as a third feed tier "New startups, no thesis evidence yet" under Main and Weak.

**Why:** the feed let in only companies that already sounded blue; Sireness (72100) never qualified (`tasks/cloud/jx28_wide_intake.md`).

**Records affected:** 50,480 companies examined, 19,027 passed the form and code test, 15,584 new, 41,220 signals; 2 of 19 cohort companies entered before cohort entry; of the other 17, 11 have no business ID and 6 have a code outside the families (46900, 03220, 47279, 58290, 70200).

**D8. 2026-10-16 23:19Z, 23:22Z, 23:33Z (17f5cee, 341ea51, fd4c16b). Stitcher limits.**

**Rule:** programme_entry, award and funding_round are emittable (the chief's decision); every other kept type stays in `stitch_log`; a second code pass re-fetches each page; stitcher events never enter the four backtest groups.

**Why:** an LLM may propose but code decides (A17).

**Records affected:** 54 companies, 175 proposed, 168 kept, 128 confirmed (refuted: 25 type not in quote, 12 company not near quote, 3 year not near quote), 29 emitted, 99 logged only; 1 of 19 cohort companies archived before the reference (VesselChain). EVAL.md also reports 27 stitcher signals in the live store; the two figures are not reconciled in the repo.

**D9. 2026-10-16 23:28Z to 23:35Z (0b3c241, a2b8eb6). The team's decision table.**

**Rule:** A12 and A11, with unknown never Low.

**Why:** the team's decision-rules doc (TASKS.md amendment 23:40Z): every company ends in exactly one of seven outcomes.

**Records affected:** every company page; investment readiness by dimension count could reach 4 of 10 at most from today's sources (a2b8eb6), so it was mostly Low or unknown until D16 changed the band.

**D10. 2026-10-16 23:35Z (a2b8eb6). Declared members get at least Medium fit.**

**Rule:** A14.

**Why:** the commit calls it a deviation from the chief's mapping; they were selected into the programme.

**Records affected:** the 19 cohort and 9 portfolio companies (28); question 01 shows the line from 02:41Z (ce502ed).

**D11. 2026-10-17 01:18Z (0cbe141). Intake families 582 and 03 added, labelled in-sample.**

**Rule:** `code_prefixes` gain 582 (software publishing) and 03 (fishing and aquaculture) in `thesis.yaml` and `baltic`, not `health`; the `tuning_note` is printed on `/method` and `/backtest`.

**Why:** the backtest showed the net missed Kilocode (58290) and Arctic Fry (03220), so their result is measured on the companies used to tune the net.

**Records affected:** intake before cohort entry went from 2 of 19 (store copy) to 4 of 19 on the live store (Kilocode, Sireness, Arctic Fry, Emission-Eye; median rank 3 of 60, EVAL.md); 16,385 `prh_intake` signals live (README.md).

**D12. 2026-10-17 02:30Z (ada70c5). Corroboration must come from a different source family.**

**Rule:** A6.

**Why:** judge round 3: Sub Marine Store's only hit was "marin" plus a PRH tax-register record from the same registry.

**Records affected:** backtest live headline 1 of 19 to 0 of 19 (0, 12, 6, 1). Feed 123 companies (70 main, 53 weak) to 122 (36 main, 86 weak); the two counts were measured before and after the rule together with the data swap (6ac05bc, README.md), so the change is not the rule alone.

**D13. 2026-10-17 02:31Z (8944ff2). Listed companies left out.**

**Rule:** A3.

**Why:** judge review round 2 item 9: a listed company is not an early signal.

**Records affected:** the live count is printed in the feed footer and `/method#left-out`; the repo states no fixed number.

**D14. 2026-10-17 02:31Z (8944ff2). Human-health practice (TOL 86) left out of `health`.**

**Rule:** A4.

**Why:** clinics matched on "medical" in the line of business; care delivery is not health technology (judge review 2 item 9, as the thesis comment cites it).

**Records affected:** `health` feed now 30 companies, 1 main, 29 weak; the health swap test moved to code 32500 (d7df036).

**D15. 2026-10-17 02:31Z (8a88193). Norway imported into the Baltic thesis only.**

**Rule:** Norwegian live-window rows enter a thesis only when NO is one of its countries; cohort matches enter every thesis.

**Why:** the commit gives none; the adapter docstring ties the live window to NO being an active country of the thesis (A15).

**Records affected:** 1,289 limited companies fetched for 17 Oct 2025 to 17 Oct 2026 (948 by NACE prefix, low; 341 by name word, medium); TuniSea found as TUNISEA AS (936277241), one exact name match (efd042d). Denmark: 1 of 6 cohort lookups answered before the daily limit (c66e1aa).

**D16. 2026-10-17 02:38Z (7413623). Bands read the page levels, and the fallback is added.**

**Rule:** the memo's bands read exactly the levels question 02 prints (TRL, BRL, IRL, Navigator category), replacing the count of investment dimensions; pass 3 adds c1 to c3 (A13).

**Why:** the memo must never say a level the page does not show (`radar/memo.py`); with fit High and impact unknown no table row can match, so every such company fell to WATCHLIST, and the fallback reads impact as implied by the sector and prints that it did (`thesis.yaml` comment).

**Records affected:** on the live copy 911 companies would read ACCELERATOR before the gate (D18).

**D17. 2026-10-16 23:04Z, 2026-10-17 00:19Z, 02:42Z (9da6080, 431df23, faf5a9b). Person names redacted in stitcher output.**

**Rule:** stored quotes, then the LLM's own logged queries, then everything the company page prints from the stitcher go through the redactor: two or more consecutive capitalised name-shaped words that are not in the legal name or the thesis business words become `[name redacted]`.

**Why:** a batch answer held a query with a private person's surname (431df23); over-redaction is the accepted cost.

**Records affected:** the stitcher block on every company page; no count of redacted strings is stated.

**D18. 2026-10-17 03:09Z and 04:35Z (79ac042, 0c19d90). The fallback gate.**

**Rule:** A13. First gate (declared member, own text, corroborated name word): 911 to 3 ACCELERATOR on the live copy. Then the registered line of business also counted as more than a code: 245 ACCELERATOR on the live store.

**Why:** a High fit that is only a coarse industry code must not carry a recommendation.

**Records affected:** 911 to 3 to 245 ACCELERATOR (the commit messages call the first a "live copy", the second the "live store"); code-only companies stay WATCHLIST.

**D19. 2026-10-17 02:00Z and 02:44Z (7667e78, 1754936). Demo examples: Viva Jets and Arctic Fry replace Kelluu.**

**Rule:** the judged documents, the search placeholder, MODULES.md and the demo path use Viva Jets Oy (2962040-6, portfolio) and Arctic Fry Oy (3166759-6, cohort).

**Why:** no commit message says why Kelluu left; DEMO_SCRIPT.md gives each company's role: Arctic Fry is the honest miss (first sales in 2020 not counted), Viva Jets has code 30120 on the list, share issues and Business Finland funding with links.

**Records affected:** README, DEMO_SCRIPT, MODULES, three templates and `tests/test_ease.py`; Kelluu stays on `/radar` and in the stitcher counts.

**D20. 2026-10-17 06:15Z (011e7ed). Consultancy codes alone demoted to weak.**

**Rule:** A7.

**Why:** such a code "is a hint that the company could be in the sector, never evidence on its own" (`code_alone_weak` comment).

**Records affected:** live-store copy, main 36 to 30 and weak 86 to 92; six companies moved: eSuunta, SU Engineering, Valavia, JSL Konsultointi, MittaTopi, ICEYE.

**Notes added in this register to D1 to D20** (the entries above are as ASSUMPTIONS.md has them):

- D1: the thesis label `readiness.business.scale` reads BRL 1-9 since 69b8e3e (2026-10-17 08:26Z, main), so the mismatch the entry notes is closed.

- D2: the reason for hiding person-shaped names is VERIFY_night2.md D-N7 ("a reviewer may read it as a named individual"); D-N1 is the reason for the CORDIS own-text part. The rule's marker words are explained in 11970db: a plain two-word name with a company suffix cannot be told from a business name.
- D12: JUDGE_REVIEW2.md item 2 counted 33 of 57 main rows resting on a same-registry corroboration; ARCHITECTURE.md read "40 of the 70 main-list companies carry the corroborated-name reason" (7667e78) and "0 of the 36" after (6ac05bc).
- D15: the reason is written at the import in `radar/adapters/brreg.py`: live-window rows belong to a thesis that covers Norway; cohort matches are imported for every thesis.

**D21. 2026-10-17 08:31Z (d4409fc, Dayo via the chief). Governance work is additive.**

**Rule:** field candidates, provenance, the stop-and-ask list and the consistency check live in new tables (`field_candidates`, `provenance`) and new files; no existing table, column, template, stylesheet or feed, memo, matching, readiness or intake rule changes; open questions mark candidate rows only (A31).

**Why:** "it must not affect what is being built" (Dayo, relayed by the chief for task jx40).

**Records affected:** no live record changed. On the blue snapshot copy, 5,357 candidate rows were written and 57 marked PENDING_DECISION; the health snapshot copy got 3,185 rows, 11 of them pending.

# T. Taxonomy

## Signal types

Every signal type a page can print, with its label from `radar/labels.py` (the only label table) and where it becomes emittable. `radar/emit.py` rejects a type that is not on its list.

| Type | Label | Emittable via |
|---|---|---|
| `register_entry` | New register entry | `radar/emit.py` |
| `bf_funding` | Business Finland funding | `radar/emit.py` |
| `eu_project` | EU project | `radar/emit.py` |
| `blueinvest` | BlueInvest pipeline | `radar/emit.py` |
| `patent` | Patent filing | `radar/emit.py` |
| `public_contract` | Public contract | `radar/emit.py` |
| `job_ad` | Open job ad | `radar/emit.py` |
| `site_customer` | Customer on own site | `radar/emit.py` |
| `first_sales` | First sales | `radar/emit.py` |
| `first_hire` | First hire | `radar/emit.py` |
| `sales_ended` | Sales ended | `radar/emit.py` |
| `employer_ended` | Employer register ended | `radar/emit.py` |
| `research_output` | Research output | `radar/emit.py` |
| `website_first_seen` | Website first seen | `radar/emit.py` |
| `website_milestone` | Website milestone | `radar/emit.py` |
| `pilot_named` | First customer or pilot named | `radar/adapters/ecosystem.py` (added at import) |
| `certification` | Certification or type approval | `radar/adapters/ecosystem.py` (added at import) |
| `programme_entry` | Accelerator or programme entry | `radar/adapters/ecosystem.py` (added at import) |
| `exhibitor` | Exhibitor or demo | `radar/adapters/ecosystem.py` (added at import) |
| `award` | Award or pitch win | `radar/adapters/ecosystem.py` (added at import) |
| `funding_round` | Funding round | `radar/adapters/ecosystem.py` (added at import) |

| Type | Label | Emittable via |
|---|---|---|
| product_launch | Product launch | radar/adapters/ecosystem.py (added at import) |
| public_funding | Grant or public funding | radar/adapters/ecosystem.py (added at import) |
| ecosystem_event | Other dated event | radar/adapters/ecosystem.py (added at import) |
| equity_event | Share issue or capital increase | radar/adapters/prh_notices.py (added at import) |
| board_change | Board change | radar/adapters/prh_notices.py (added at import) |
| capital_loss | Loss of share capital | radar/adapters/prh_notices.py (added at import) |
| startup_notice | Start-up notification | radar/adapters/prh_notices.py (added at import) |

eu_grant has a label and is a counted type in the thesis files, but it is not on the emit list, so radar/adapters/eu_grants.py writes eu_project instead (its own comment says the add is one line in radar/emit.py, a contract file the chief owns).

### Sources and source families

| Adapter | Label on pages | Source family (A6) |
|---|---|---|
| blueinvest | BlueInvest pipeline | blueinvest |
| brreg | none in radar/labels.py (falls back to the plain id) | brreg |
| businessfinland | Business Finland export | businessfinland |
| cordis | CORDIS | cordis |
| cvr | none in radar/labels.py (falls back to the plain id) | cvr |
| declared | declared list | declared |
| ecosystem | declared research | ecosystem |
| estonia_register | Estonian e-Business Register | estonia_register |
| eu_grants | EU grants | eu_grants |
| hilma | Hilma notice | hilma |
| openalex | OpenAlex | openalex |
| prh | PRH/YTJ register | prh |
| prh_intake | PRH/YTJ register (wide intake) | prh |
| prh_notices | PRH registered notices | prh |
| prh_recent | PRH/YTJ register (recent registrations) | prh |
| stitcher | AI stitcher (checked by code) | stitcher |
| tax_registers | PRH/YTJ tax registers | prh |
| ted | TED contract awards | ted |
| webarchive | Internet Archive | webarchive |

brreg and cvr have no entry in radar/labels.py; pages fall back to the plain id. Proposed labels: "Norwegian register (Brønnøysund)" and "Danish CVR register".

### Confidence levels (A9)

| Level | Rule (radar/emit.py) |
|---|---|
| high | the source names the company or its business ID directly and states the date |

| Level | Rule (`radar/emit.py`) |
|---|---|
| medium | matched by an exact legal name or website domain, date stated by the source |
| low | matched by name keyword or industry code only; shown, never counted in rankings |

### Verdicts (A12, A13)

Seven outcomes, the team's decision table in `memo.decisions` of `config/thesis.yaml`: INVEST, DUE DILIGENCE, ACCELERATOR, GRANT SUPPORT, PARTNER INTRODUCTION, WATCHLIST, REJECT / LOW FIT. No match is WATCHLIST. Levels are banded Low / Medium / High / unknown (A11); unknown never matches a cell that needs a band.

### Group tags

| Tag | Meaning | Written by |
|---|---|---|
| `cohort` | one of the 19 Nordic Blue cohort companies | `radar/adapters/declared.py` |
| `portfolio` | one of the 9 Redstone portfolio companies | `radar/adapters/declared.py` |
| `feed` | found by a public source; may enter the early-signal feed | discovery adapters |
| `intake` | wide intake, startup-shaped with one startup signal (A2) | `radar/adapters/prh_intake.py` |
| `ecosystem` | inserted from the declared ecosystem dataset, outside the feed | `radar/adapters/ecosystem.py` |

`cohort` and `portfolio` are never downgraded to `feed` (`emit.upsert_company`).

### Field candidate statuses (A28, A31)

`applied` (the value in the companies table, what live pages show), `agrees`, `differs`, `not_comparable` (two languages), `store_empty` (the store holds no value), `PENDING_DECISION` (an open question names the row; never on an applied row).

## F. Formats

| What | Format | Where it is checked |
|---|---|---|
| Event and founding dates | ISO 8601 calendar date `YYYY-MM-DD`, the day the event happened, never the fetch date (A18) | `radar/emit.py`, `radar/formats.py` `date_ok` |
| Date precision | `day` unless `signal_meta.date_precision` says `year` (stored as 1 January, printed "2026 (year only)") or `month` (stored as the last day of the month, labelled); a field candidate carries `day`, `month` or `year` | `radar/schema/005_signal_meta.sql`, `field_candidates.precision` |
| Times | UTC, `YYYY-MM-DDTHH:MM:SSZ` (`fetched_at`, `built_at`, provenance `at`) | `radar/emit.py` `now_utc` |
| Business ID (Y-tunnus) | `NNNNNNN-C`; C is the check digit: weights 7, 9, 10, 5, 8, 4, 2; remainder 0 gives 0, remainder 1 is never issued, otherwise 11 minus the remainder | `radar/formats.py` `business_id_ok` |
| Country | ISO 3166-1 alpha-2, upper case (A29) | `radar/formats.py` `country_ok` |
| Amounts | `amount_eur`: euros as the source states them, no conversion and no exchange-rate code; NULL when the source states none (A19) | `radar/schema/001_core.sql` |
| Money and dates in Finnish copy | Finnish format (CLAUDE.md), no em dashes | page templates |

| What | Format | Where it is checked |
|---|---|---|
| Links | every signal and candidate carries an `http(s)` link a human can open; nothing behind a login (A27) | `radar/emit.py`, `field_candidates` CHECK |
| Rule ids | A<n> (assumption) and D<n> (decision) in this file; question ids B<n>-NN in `QUESTIONS_B<n>.md` | `radar/consistency.py`, `provenance` CHECK |

## Q. Open questions

Batch B1 (`QUESTIONS_B1.md`, 17 Oct): 15 issues. B1-01 to B1-11 come from the field candidate conflicts, and B1-12 to B1-15 from the stop-and-ask scans: duplicates, dates, amounts. Until they are answered, live pages keep A1. When answers land they become D22 onwards, and this file is re-issued.

## K. Known inconsistencies between documents and code (for the chief; nothing changed here)

- `config/thesis.yaml:26` (and the Baltic file) still says "TOL 2008 (NACE) codes" above a list that D6 moved to TOL 2025 codes.
- ARCHITECTURE.md (the "with a week" list) still proposes the source-family corroboration rule that D12 implemented.
- EVAL.md (health section) still says the health feed lists clinics matched on "medical"; D14 left TOL 86 out. It may refer to clinics coded outside 86; the repo does not say.
- D8: EVAL.md reports 27 stitcher signals in the live store and the batch reports 29 emitted; the repo does not reconcile the two.
- `eu_grant` is labelled and counted but not emittable (see T).

## H. History of this register

- 2026-10-17 08:19Z: ASSUMPTIONS.md, A1 to A27 and D1 to D20, drafted on the VPS (bf4547c); merged to main as e0568b2.
- 2026-10-17: folded into this file with the same ids (task jx40). Added A28 to A33, notes to D1, D2, D12 and D15, D21, the taxonomy, the formats and batch B1.